INCH-POUND

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MILITARY SPECIFICATION

SERVO DRIVE, ELEVATION: 10548062

This specification is approved for use within the U. S. Army **Armament, Munitions and chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.**

1. **SCOPE**

**1.1 Scope.** This specification covers the elevation servo drive which is a part of the Antenna Assembly AS-2049/VPS-2 used with Radar Set AN/VPS-2 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control, Materiel, General Specification Governing the Manufacture and Inspection of
MIL-S-20708 - Synchos, General Specification for

**Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.**

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MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-S-45743 - Soldering, Manual Type, High Reliability, Electrical and Electronic Equipment
MIL-S-46844 - Soldering, Bath Soldering of Printed Wiring Assemblies
MIL-A-50687 - Antenna Assembly AS-2049/VPS-2: 10556312

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Military Specifications and Standards, Bldg. 4X), 700 Robbins Avenue, Philadelphia, PA 19111-5094. )

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548062 Elevation Servo Drive
10548064 Elevation Servo Drive Runs List, Wire
10548065 ***Housing,*** Elevation Drive
10548069 Trunnion, Elevation
10548193 Servo Electronics
10548389 Antenna Schematic
10556123 Clamp Assembly, Shaft
10556312 Antenna Assembly, AS-2049/VPS-2

PACKAGING DATA SHEET

PS10548062 - Packaging of Elevation Servo Drive



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(Copies of drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity. )

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein (except for associated-detail specifications, specification sheets or MS standards) , the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accorcdance with tne applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10548062, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the elevation servo drive shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.

b. Inorganic protective surface finishings.

c. Part identification and marking.

d. workmanship.

e. Soldering electrical connections.

3.3 Environmental service conditions.

3.3.1 Vibration. The elevation servo drive shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in-each of three mutually perpendicular axes, as follows:

| Frequency Range | Amplitude |
|---|---|
| 5 to 15 hertz | 0.06 inches double amplitude |
| 15 to 25 hertz | 0.04 inches double amplitude |
| 25 to 55 hertz | 0.02 inches double amplitude |

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The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 5-55-5 hertz sweep time is 7 minutes 35 seconds ± 10%.

**3.3.2 Shock.** The elevation servo drive shall not suffer damage and shall meet requirements of 3.4 **and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock Impulses of 15 g's ± l0%, 11** milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

**3.3.3 Altitude.**

**3.3.3.1 Operating. The elevation servo drive shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).**

3.3.3.2 Non-operating. The elevation servo drive shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

**3.3.4.1 Operating. The elevation servo drive shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 deg. F and +155 deg. F.**

3.3.4.2 Non-operating (storage). The **elevation *servo* drive shall not suffer permanent damage when subjected to the constant temperature** extremes of -60 deg. F for a period of 24 hours and +160 deg. F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

**3.3.5** Humidity (non-operating) The elevation servo drive **shall meet the requirements of 3.4 and 3.5** after exposure to an atmosphere with a-relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous **as** specified on the schematic diagram of the elevation servo drive.

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3.5 Performance.

3.5.1 Component resistance. After 10 minutes of operation, the resistance of components with traveling elements to fixed elements shall be **as follows:**

**a.** Drive, motor resistance: 0.75 to 10 ohms.
b. Tachometer generator: 350 ± 150 ohms.
**c.** Synchro stator S1 to S2 or S3: 500 ± 100 ohms.
d. Synchro rotor R1 to R2: 320 ± 80 ohms.
e. Drive brake: 200 ± so ohms with diode cathode connected to terminal J1-n.

**3.5.2** Servo drive brake. When +26 ± 2 volts (V) is applied between J1-n and J1-a, the servo drive brake shall release.

3.5.3 Drive motor. When conditions of 3.5.2 are present, the drive motor shall begin to operate between 0.3 and 3.o volts direct current (Vdc) of either polarity.

3.5.4 Tachometer. When the drive voltage of 3.5.3 is increased to 10 V the tachometer (generator) shall produce 13.0 ± 4.0 V and **shall be** steady within 1.0 V.

3.5.5 Synchro. The synchro (B1) shall be aligned to electrical zero to within 1/8 deg.

3.5.6 Potentiometer. The \ potentiometer shall provide a null of less than 120 millivolts (mV) at O deg. elevation lead angle and shall provide 2.4 ± 0.4 V at ± 10 deg., when excited with 15 Vdc. and potentiometers shall provide a positive change in voltage when rotation is from **-25 deg. to +25** deg. elevation.

3.5.7 Limit switch. A limit switch shall operate at +97 ± 2 deg. elevation and again operate at -10 ± 2 deg. below horizon.

**3.5.8 Gearing. There** shall be no evidence of backlash in the data train gearing.



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3.5.9 Servo drive dynamic test. The elevation servo drive after having met the requirements of 3.5 through 3.5.8 shall meet the dynamic test as specified in MIL-A-50687 when substituted into an antenna AS-2049/VPS-2. Antenna AS-2049/VPS-2 and test facilities shall be as specified in MIL-A-50687. When an elevation servo drive is procured as a spare item, the antenna **assembly (10556312)** and servo electronics (10548193), referenced in MIL-A-50687, will be supplied to the contractor **as** GFP.

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Workmanship. The workmanship requirements of MIL-F-13926, MIL-S-45743 and MIL-S-46844 shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except **as** otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define quality assurance terms herein.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections **3 and 5** and all applicable drawings. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification.



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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

| | | |
|---|---|---|
| a. | Temperature | +50° F to +90° F |
| b. | Relative Humidity | 90% RH Max. |
| c. | Pressure | 28 to 32 inches of Hg |

4*3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample **as** designated by the contracting officer consisting of three (3) elevation servo drive for inspection in accordance with 4.3.2.

**4.3.2 Inspections to be performed.** First article assemblies, components and test specimens may be subjected by the Government to any of all the examinations and tests specified in Table I and Table II herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to **comply with any of the applicable requirements, the first article sample shall be rejected.** The Government reserves the right to terminate its inspection upon failure of any assembly, component, or test specimen.



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4.4 Quality conformance inspection.

**4.4.1** Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from **which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to** represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and the same specification revision and complies with - the provisions for submission of product. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examination and tests.

4.4.2.1 Components, assemblies and **subassemblies. All** components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the technical data package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table I shall **be** performed on a 100% basis.



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TABLE 1. Functional tests (loo percent inspection) .

| CHARACTERISTIC | | REQUIREMENT | TEST PROCEDURES |
|---|---|---|---|
| CRITICAL: | | None | |
| MAJOR : | | | |
| 1 0 1 | **Electrical** | 3.4 | **4.7.2** |
| 102 | Performance | 3.5 | **4.7.3** |
| MINOR : | | **None** | |

4.5 Special sampling. One elevation servo drive out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table I and shall then meet the requirements and tests in Table II. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE II. Special sampling, environmental.

| CHARACTERISTIC | | REQUIREMENT | TEST PROCEDURES |
|---|---|---|---|
| CRITICAL: | | None | |
| **MAJOR** : | | | |
| 1 0 3 | Vibration | **3.3.1** | 4.7.1.1 |
| **104** | Shock | **3.3.2** | 4.7.1.2 |
| **105** | Altitude | **3.3.3** | 4.7.1.3 |
| **106** | Temperature | **3.3.4** | 4.7.1.4 |
| **107** | Humidity (nonoperating) | **3.3.5** | 4.7.1.5 |
| MINOR : | | **None** | |

**4.5.1 Failure of sample. Should any one item** of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the



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**contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.**

4.6.2 Contractor-furnished inspection equipment.

**4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.**

**4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of** Inspection" specified in 4.7 of-this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming.
Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the standard test equipment listed in Table III shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE 111. Test equipment list.

| Item Number | Description |
|---|---|
| 1. Servo drive tester | **See Figure 1. See Table IV for parts list.** |
| 2. DC power supply | **24 Vdc, 0.5** A.<br>**Load regulation ± 1%.**<br>Ripple: Less than 100 mV. |



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TABLE 111. Test equipment **list (continued )**

| | | |
|---|---|---|
| 3. | DC power supply | **O to 30 Vdc, 10 A.**<br>**Load regulation:** ± 1%.<br>**Ripple: Less than** 100 mV. |
| 4. | **DC** power supply | 15 **Vdc, 100 mA.**<br>Load regulation: ± 0.1%.<br>Ripple: Less than 10 mV. |
| 5. | Aligning fixture for elevation assembly | See Figure 2. |
| 6. | Oscilloscope | Sensitivity: 0.1 V/cm.<br>Frequency response: 100 kHz.<br>Time **base: 1 to 100 ms/cm.** |
| 7. | AC-DC voltmeter | Range: 500 mV to 150 v.<br>Input impedance: 10 megohms. |
| 8. | Shaft clamp | 10556123 |
| 9. | Inspection mirror | Lighted, flexible inspection mirror. |
| 10. | Ohmmeter | R x 1 scale: O to 1 k Ohm.<br>R x 100 scale: O to 100 k ohm.<br>Short test current R x 1 scale: Not **more** than 200 mA.<br>Short test current R x 100 scale: Not more than **20 rnA.** |

TABLE IV. Servo drive test harness parts list.

| Designation | Nomenclature | Part number |
|---|---|---|
| **S1** | **SPST toggle switch** | *MS35058-22* |
| **TP1-TP24** | **Banana jack** | **H. H.** Smith DF30 |
| **B1** | **Control transmitter** synchro | 23CX4 with Theta Instruments, Model DA-23-2405 dial assembly or equivalent. |
| R1 | **Resistor** | RCO7GF1O3J |



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| | | |
|---|---|---|
| cl | Capacitor | TRW capacitor X663F, 1 ~F, 100 WVdc or equivalent. |
| J1 | **Connector** | Bendix PT06A-20-39S or equivalent. |
| DS1 | **Pilot lamp** | MS25257-4 |



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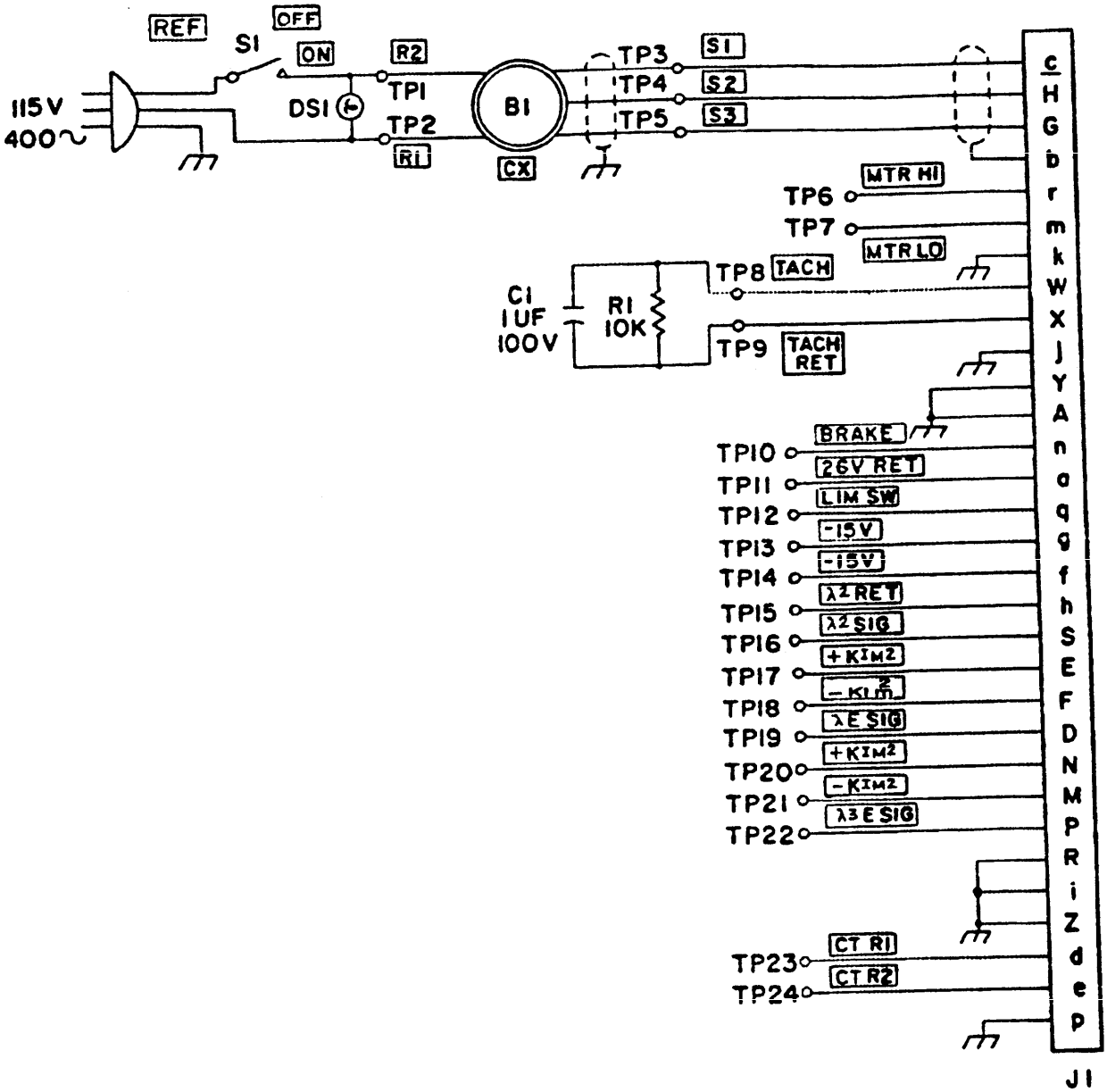



FIGURE 1. Elevation servo drive tester.



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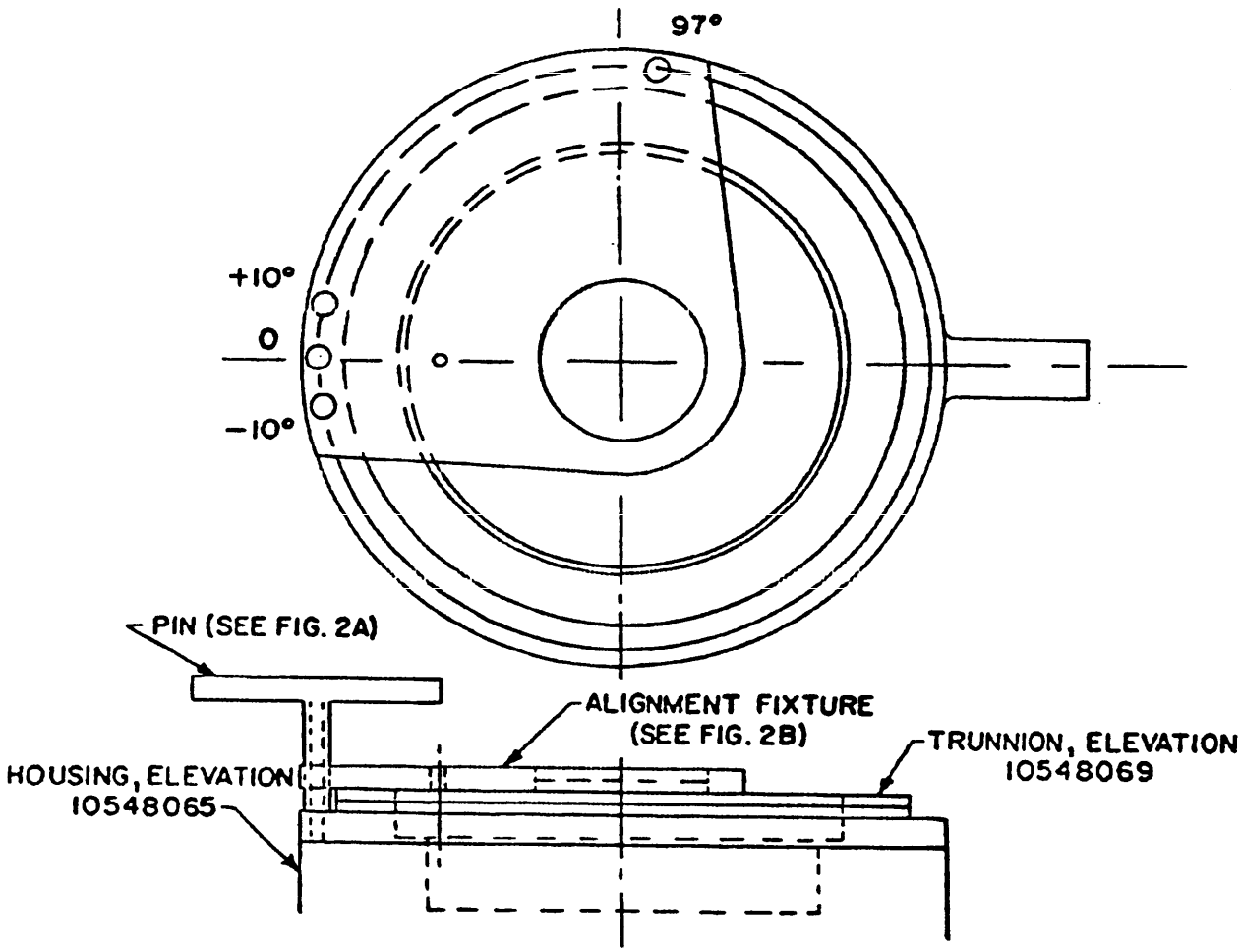



FIGURE 2. Aligning fixture for elevation assembly.



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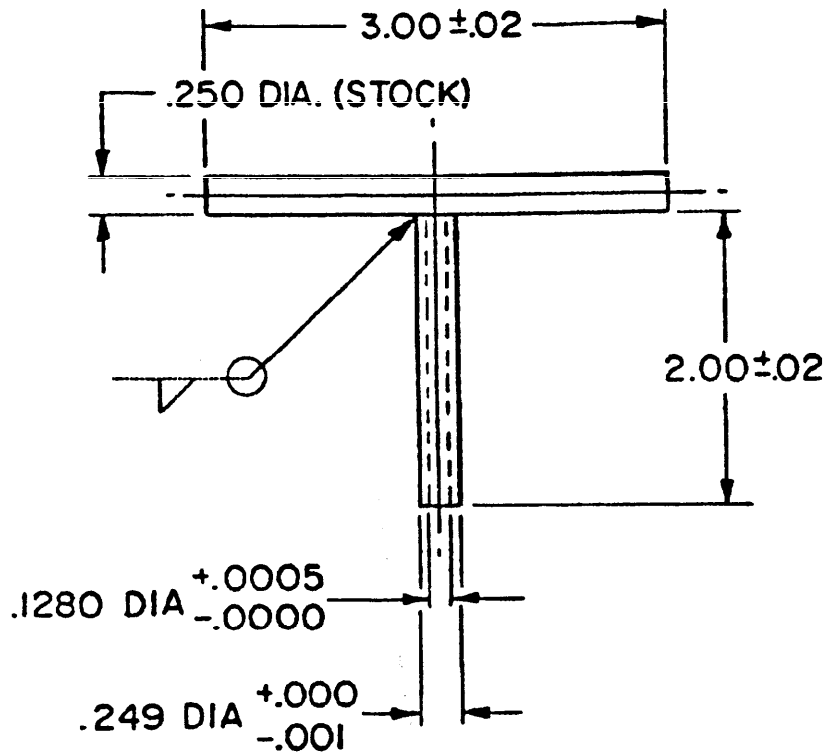



FIGURE 2A. Pin aligning fixture.



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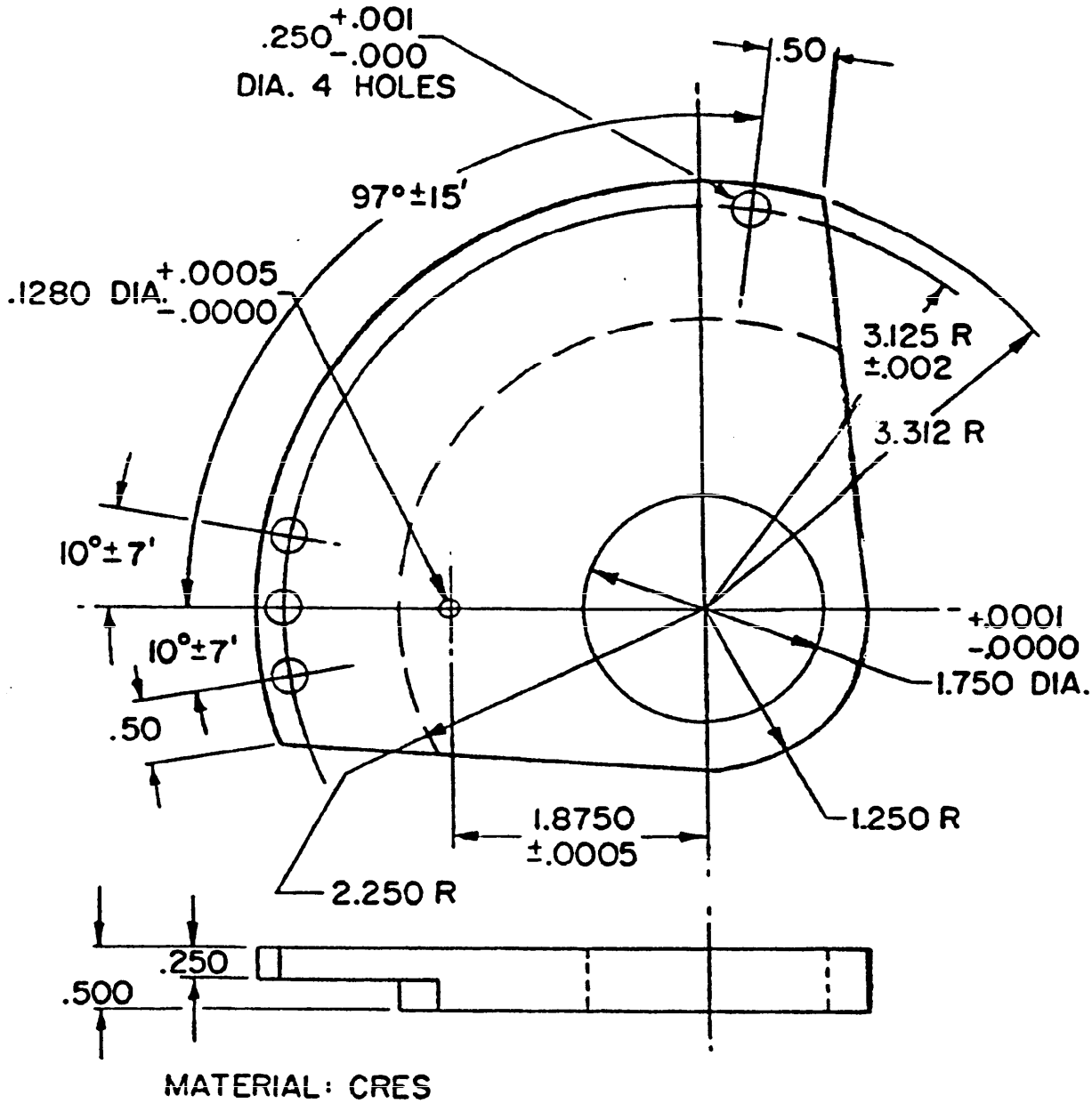



FIGURE 2B. Elevation aligning fixture.



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4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test **shall be performed as specified in Method 514.3, Category 8, procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I, only steps 4 through 8 of paragraph 11-2.1 apply. The elevation servo drive shall be attached to the vibration generator directly or with a fixture, and securely** held by its normal means of attachment. After vibration, the elevation servo drive shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.l.

4.7.1.2 Shock . This test shall be performed as specified in Procedure I, Section II of Method 516.3 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

**b. The** shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the elevation servo drive shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating) This test shall be performed as specified in Procedures II and III, Method 500.2, of MIL-STD-810. The operation checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.



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4.7.1.4 Temperature.

**4.7.1.4.1 High (operating and non-operating). This test shall be Performed as specified in Procedures I and II, Method 501.2, of** MIL-STD-810 with the constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 LOW (operating and non-operating) . This test shall be performed as specified in Procedures I and II. Method 502.2, of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

**4.7.1.5** Humidity (non-operating) This test shall be performed in accordance with Procedure III, Method 507.2, of MIL-STD-810 except that in **Step 7,** checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with **3.3.5.**

4.7.2 Electrical. Check the wiring visually using schematic 10548389 and runs list 10548064 as references. Particularly check the **color of the leads. This test** shall be performed only at the beginning and end of the performance testing.

> CAUTION:When measuring potentiometers with an ohmmeter, set ohmmeter to X 100 scale to start, and then switch to Xl scale after a short is measured on X 100 scale. This will prevent damage to the function potentiometers.

4.7.3 Performance. The selected elevation servo drives shall be tested as follows:

4.7.3.1 Brake activation. Check that REF switch on servo drive tester is in the OFF position. Connect the test cable between the tester and the unit under test. Connect power supply, item 2, with positive lead to BRAKE (TP10) and negative lead to 26 V RET (TP1l). Adjust for 24 V. Listen for brake actuation.

4.7.3.2 Motor run-in. Connect servo drive tester to elevation servo drive. Connect a dc power supply, item 3, with positive terminal connected to MTR HI (TP6) and-negative terminal connected to MTR LO (TP7). Turn on power supply and increase voltage until drive starts turning. Increase voltage to 10 V and allow motor to run in alternating rotation directions for l0 minutes. Then, turn off the power supply and disconnect the test leads.



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4.7.3.3 Resistance check.

a. J1-r to J1-m: 0.75 to 10 ohms.

b. J1-r to J1-Y: Open.

c* J1-W to J1-X: 200 to 500 ohms.

d. J1- c to J1-H, J1- c to J1–G and J1-G to J1-H: 400 Ohms to 600 Ohms.

e. **J1-d to J1-e: 240 to 400 Ohms.**

f. **J1-Y to chassis: Short.**

g. **J1-A to chassis: Short.**

h. **J1-n to J1–a (with positive voltage on n): 125 to 275 ohms.**

i. **J1-n to J1-a (with positive voltage on a):** Less than 100 ohms on Xl ohmmeter scale.

4.7.3.4 Motor-generator.

a. Connect the dc power supply, item 3, with positive terminal connected to MTR HI (TP6) and negative terminal connected to MTR LO (TP7). Turn on output voltage supply. Increase voltage until drive starts turning. Applied voltage should be between 0.3 and 3 V. The drive should rotate clockwise with respect to the housing. Connect a VOM set to read dc voltage with negative lead on TACH (TP8) and positive lead on TACH RET (TP9).

b. Increase motor voltage to 10 V. Tachometer voltage shall be 9 to 17 v, of the polarity indicated in a.

c. Tachometer voltage shall be steady within 1 V as the unit is run.

d. Turn off power supply. Reduce output voltage control to zero. Reverse power supply polarity. Turn on supply and increase voltage until drive starts turning. Applied voltage should be between 0.3 and 3 V.



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e. The drive should rotate counterclockwise. Increase the motor voltage to 10V. Reverse tachometer leads to obtain +9 to +17 V at TACH (TP8) with respect to TACH RET (TP9). Tachometer voltage shall be steady within 1 V as the unit is run. Turn off the motor power supply and disconnect the VOM.

4.7.3.5 **Synchro.**

a. **Install elevation assembly alignment fixture (see figure 2) and move drive to 0 deg., using motor power supply. Adjust CX dial for exactly 0 deg. Position gun elevation input shaft at approximately midrange of rotation and hold in place using the clamp shown in Figure 3. Slide shaft clamp, item 8, onto gun elevation input shaft, then slide the mechanical stop pin into alignment hole of shaft clamp assembly. Tighten setscrews of shaft clamp assembly to fasten shaft clamp assembly to gun elevation input shaft and mechanical stop pin. This** fixture maintains **gun elevation input shaft at approximately 0 deg. elevation.**

NOTE: CX of servo drive tester to **be zeroed in accordance with** MIL-S-20708.

Turn gun elevation input shaft until elevation limit pin is approximately 180 deg. from mechanical pin extending from elevation servo drive shown in Figure 4, onto gun elevation input shaft, then slide the mechanical pin into alignment hole of shaft clamp assembly. Tighten three setscrews of shaft clamp assembly to fasten shaft clamp assembly to shaft. This fixture maintains gun elevation input shaft at approximately 0 deg. elevation.



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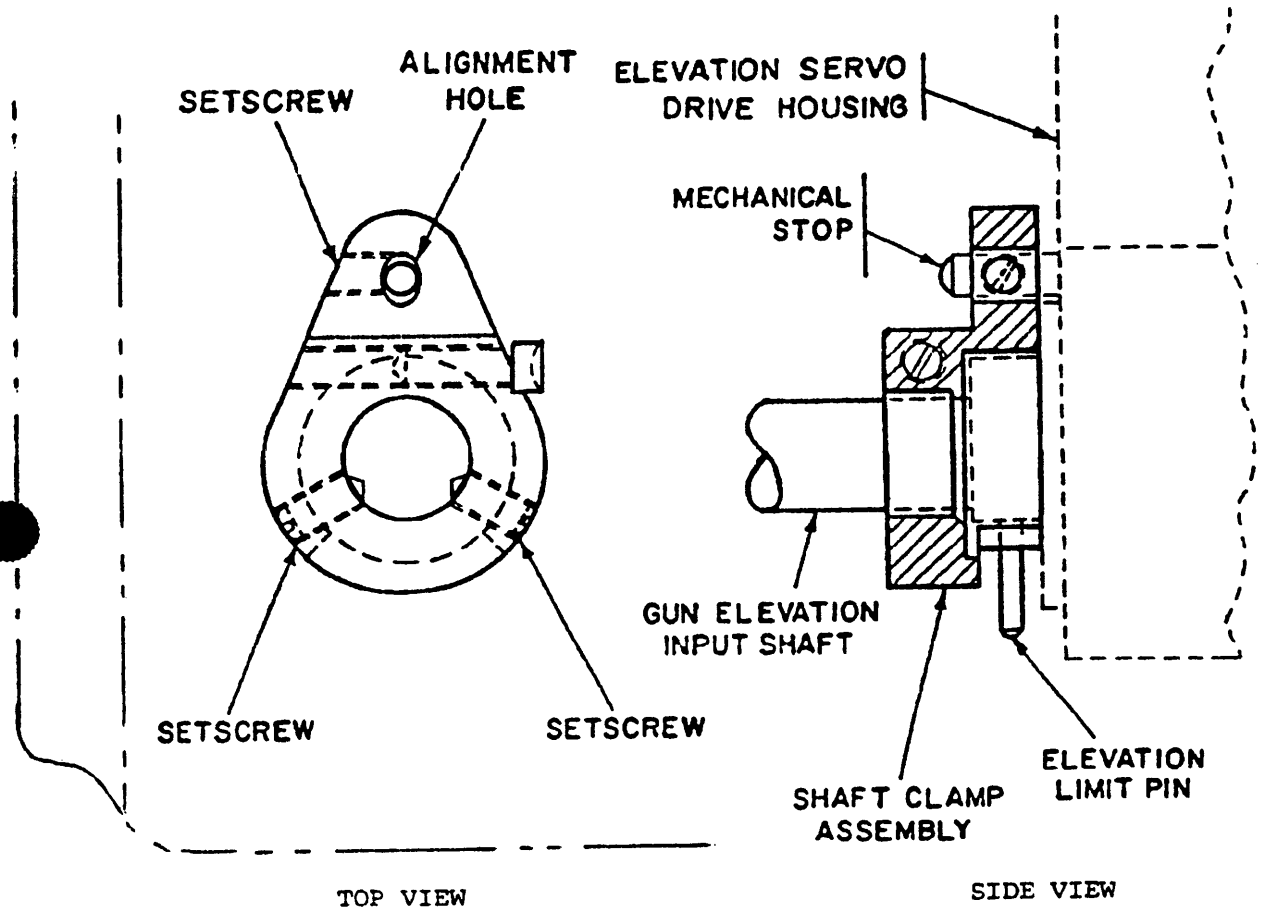
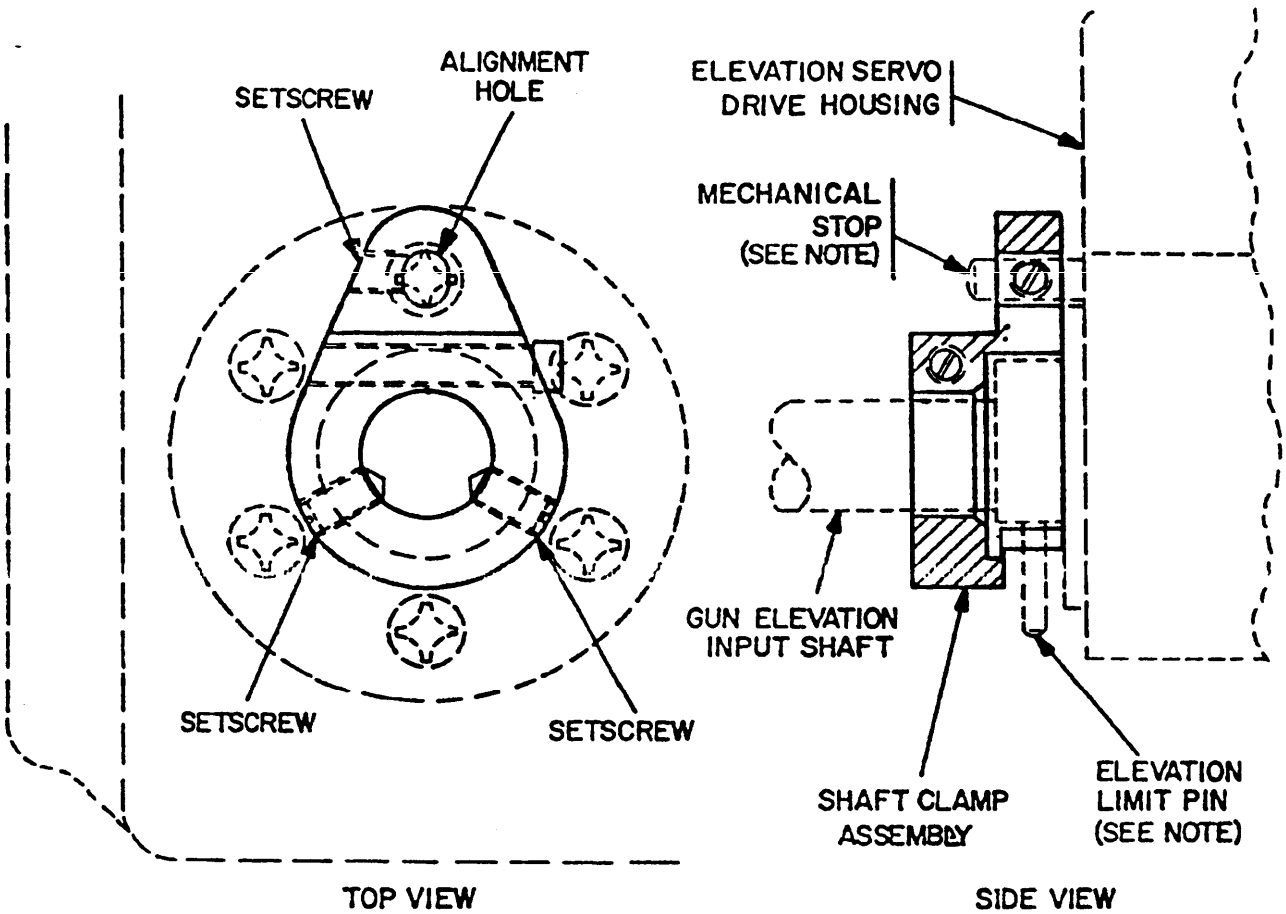



FIGURE 3. Elevation assembly alignment fixture operation.



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NOTE: THE ELEVATION LIMIT PIN AND THE MECHANICAL STOP PREVENT THE ELEVATION INPUT SHAFT FROM ROTATING MORE THAN 360°.

FIGURE 4. Shaft clamp assembly placement.



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b. Connect a jumper from CX R1 (TP2) to CT R1 (TP23) and from CX R1 (TP2) to CX S2 (TP4). Connect a floated oscilloscope with the single lead at CT R2 (TP24) and the return at CT R1 (TP23).

c. Rotate the motor drive around 0 deg. by rotating the slip clutch gear by hand until a null is obtained on-the oscilloscope. Connect the second channel of the oscilloscope to S1 (TP3). Display both waveforms. Rotate the drive in such a way as to produce a counterclockwise rotation of the synchro shaft (as seen by looking at the shaft end). The two waveforms should appear in phase. Using an inspection mirror verify that the mark on the synchro shaft is in alignment with the mark on the synchro housing (this will assure proper synchro zero). Set REF switch to **OFF.** Disconnect jumpers (except brake power supply) and oscilloscope.

**4.7.3.6 Function potentiometer ,**

**a.** Position the drive to the 10 deg. **above** zero position by releasing the brake and using the drive motor. Use the **aligning-fixture of Figure 2 to indicate to +10 deg. position. Set the synchro of the servo drive tester set to 12.25 deg.**

CAUTION : Always approach from null. Set brake power supply to OFF. Connect a jumper between TP13 and TP14 (-15v). Connect dc power supply, item 4, to provide -15 V at TP13 with respect to > RET (TP15). Connect a dc null voltmeter between $A_{2}$ SIG (TP16), and $A_{2}$ RET (TP15) and read a voltage of 2.4 + 0.4 V. Position the drive to the 10 deg. below zero position by releasing the brake and using the drive motor. Use the aligning fixture of Figure 2 to indicate the -10 deg. position. Set the synchro of the servo drive tester to 347.75 deg. (-12.25 deg.).

CAUT0N : Always approach from null. The voltage at this 10 deg. position shall also be 2.4 ± 0.4 V. In addition, the difference between this voltage and the voltage obtained at the other 10 deg. position shall not differ by more than 0.2 V.



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b. Remove **the** alignment pin and rotate the drive to the O deg. position (the null position). A null voltage no greater than 120 mv shall be obtained.

c. Rotate the drive slowly while observing the voltmeter. The voltage should vary smoothly from null to -15 V and then drop suddenly back to zero at about -25 deg. The same response should be seen when rotating the drive in the opposite direction. Turn off the -15 V supply and disconnect it. Remove the jumper between TP13 and TP14.

d. Connect the same supply with +15 V at TP17 with respect to the TP18 and connect high impedance voltmeter to E SIG TP19) with respect to TP18. As the drive is rotated from about +25 deg. to -25 deg. , the voltage will drop to zero again. Below -25 deg. , the voltage will jump to +15 v. Turn off 15 V power supply and disconnect it as well as the voltmeter.

e. Connect this power supply with +15 V at TP20 with respect to TP21; connect a high impedance voltmeter at ~[3]E SIG (TP22) with respect to TP21. As the drive is rotated from about -25 deg. to 25 deg. , the voltage should increase smoothly from O to **+15 V. Below -25 deg. the** voltage will jump to +15 V again. Above +25 deg. , ' the voltage will jump to O V. Turn off 15 V power supply and disconnect it as well as the voltmeter.

**4.7.3.7** Microswitch stop.

a. Connect an ohmmeter between TP11 and TP12. Move elevation servo drive to +90 deg. on elevation alignment fixture. Continue to move elevation servo drive until ohmmeter shows microswitch operation. Stop drive **at** this position. Aligning fixture pin shall be capable of being inserted into 97 deg. hole to provide locking. Remove aligning fixture pin.



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b. Move elevation servo drive to 0 deg. Continue to move elevation servo drive until ohmmeter shows microswitch operation. Stop drive at this position. Aligning fixture pin shall be capable of being inserted into -10 deg. hole to provide locking. Remove aligning fixture pin.

4.7.3.8 Backlash.

a. All antibacklash **gears shall be visually inspected using item 9 and shall be loaded (spring, compressed) between two to four teeth. This will provide approximately 10 inch-ounces of preload in the data gear train.**

b. When the gear train is deflected to overcome the antibacklash gears **and is** released the gears shall return to their original position with no evidence of loose gears or mechanical play.

4.7.3.9 Servo drive dynamic test. The elevation servo drive after having met the requirements of **3.5.3** through 3.5.3.8 shall meet the dynamic test as specified in MIL-A-50687 at ambient conditions when substituted into an Antenna AS-2049 VPS-2. Antenna AS-2049 VPS-2 and test facilities shall be as specified in MIL-A-50687 (see 3.5.9).

5. PACKAGING

5.1 General. Preservation, packing and marking shall **be in** accordance with Packaging **Data Sheet P510548062.**

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.

6.1 Intended use. The elevation servo drive provides mechanical movement of the antenna in elevation when used with Radar Set AN/VPS-2. The potentiometer provides lead angle information to the sight current generator, required when solving the fire control problem.



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6.2 Acquisition requirements. Procurement documents should specify the following:

**6.2.1** Procurement requirements.

a. Title, number and date of this specification.

b. Applicable stock number.

**c*** First article sample requirements. .

d. Packaging requirements, if other than specified in Section 5.

**e.** Serialization requirements, if applicable.

f. Certificate of Conformance for each lot or shipment of product.

6.2.2 Submission of inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-1, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

**6.3 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command** (ARRADCOM) , Edgewood **Arsenal,** Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now unde~- **cognizance** of ARDEC.

6.4 Subject term (Key word) listing.

Radar Set AN/VPS-2
Elevation Servo Drive
Elevation Assembly



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6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
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Preparing activity:
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| OMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
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| | MIL-S-5-667A (AR) | 16 MAY 1990 |

IENT TITLE

SERVO DRIVE, ELEVATION: 10548062

OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

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